

Trainings ▾

Select Service Tools

Recommended Cummins® Service Tools

- Assembly lubricant, Part Number 3163087, or equivalent.

Additional Service Items

- No additional service items required.

Preparatory Steps

WARNING

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

WARNING

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

WARNING

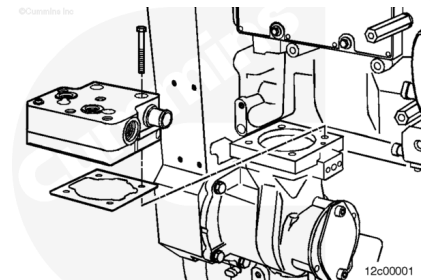
Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

- Disconnect the batteries. See equipment manufacturer service information.
- Drain the cooling system. Refer to Procedure 008-018 in Section 8. (</qs3/pubsys2/xml/en/procedures/56/56-008-018-tr.html>)
- Remove the air inlet and outlet connections from the air compressor. Refer to Procedure 012-014 in Section 12. (</qs3/pubsys2/xml/en/procedures/56/56-012-014-tr.html>)
- Remove the air governor or air governor signal line, if equipped. See equipment manufacturer service information.

Remove

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



Remove the four air compressor head mounting capscrews.

Discard the capscrews.

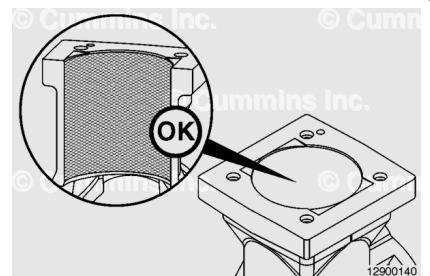
Note : Record the position of the intake and exhaust ports in the air compressor cylinder head with respect to the crankcase for proper assembly.

Remove the air compressor head and gasket.

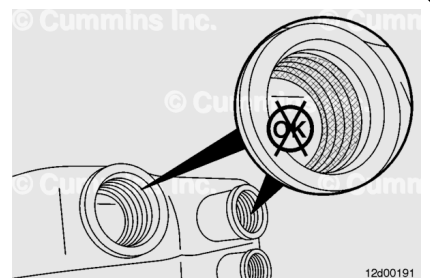
Discard the gasket.

Clean and Inspect for Reuse

Inspect the air compressor cylinder bore for damage. Refer to Procedure 012-014 in Section 12.
(/qs3/pubsys2/xml/en/procedures/56/56-012-014-tr.html)

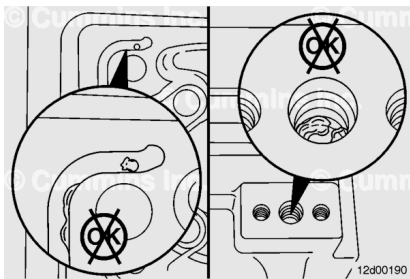


Inspect the intake and exhaust ports for excessive carbon buildup. Refer to Procedure 012-003 in Section 12.
(/qs3/pubsys2/xml/en/procedures/56/56-012-003.html)



Inspect the air governor signal channel in the air compressor head. The channel should appear as a clean aluminum surface. If there is any sign of rust, dirt, or exhaust gas recirculation debris, the air compressor head **must** be replaced.

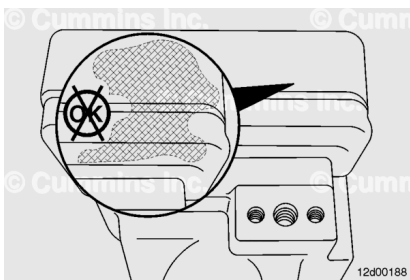
Note : If rust is found in the governor signal channel, it is the result of poor system maintenance. Check for proper operation of the air dryer and the wet tanks. See equipment manufacturer service information.



Inspect the cylinder head for signs of a coolant leak.

A coolant leak will be identified by an area of engine paint appearing to have been washed away. White crystalline clusters of material are **not** a sign of a coolant leak.

If a leak is found, replace the air compressor cylinder head.

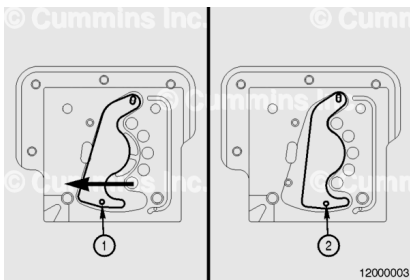


Verify the unloader slider plate is fully closed when at rest.

Remove the unloader slider plate and verify proper unloader piston travel. Use a stiff wooden handle or equivalent non-abrasive device to move the unloader piston by pushing back on the unloader slider plate guide pin in the direction of arrow as indicated. Be careful **not** to damage the aluminum surface of the head or drop debris into the open slot in which the guide pin travels.

Note : When released, the unloader piston should return to the loaded position with a quick, smooth motion and an audible snap.

If the unloader piston does **not** return to the loaded position with a quick, smooth motion and an audible snap, reference the table below for repair direction:



Guide Pin Action	Repair Direction
Guide pin can be moved to loaded (closed) position by hand.	Spring is broken or weak; replace cylinder head.
Guide pin does not return or slow to return to loaded position.	Refer to Procedure 012-003 in Section 12. (/qs3/pubsys2/xml/en/procedures/56/56-012-003.html)

Assemble

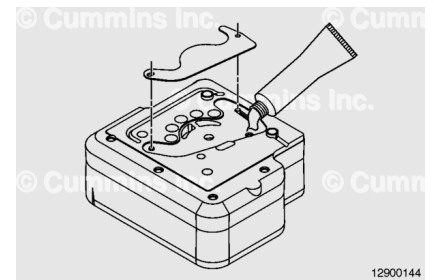
Note : The assembly process is **not** required for new air compressor cylinder head kits. These kits include a cylinder head with all of the gaskets assembled and ready for installation on the air compressor crankcase.

Note : Make sure the unloader slider plate, gasket surfaces and valve plate are free of any debris before assembling the unloader slider plate.

Apply a spot of assembly lubricant, Part Number 3163087, or equivalent, to the back of the unloader slider plate and to the unloader slider plate guide pin.

Install the unloader slider plate onto the air compressor head by locating it on the unloader slider plate pivot pin and the unloader slider plate guide pin.

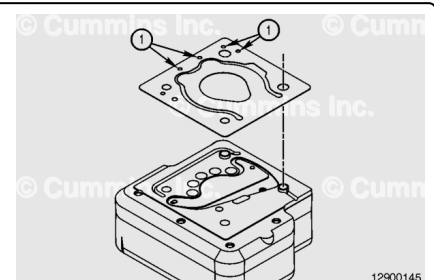
Note : Make sure the unloader slider plate is properly seated and that enough assembly lubricant has been used to securely hold it in position during the installation procedure or damage to the unloader slider plate can occur.



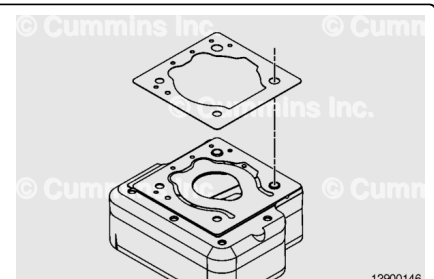
Make sure the intake valve is oriented so the governor signal channel of the cylinder head is exposed by four holes (1).

Install the air compressor intake reed valve. The intake reed valve has larger diameter holes on one side than the other to assist in proper orientation. The larger diameter hole is to be located on the spring pin protruding from the bottom of the air compressor cylinder head. When properly installed, the free end of the intake reed valve will cover the five circular intake holes in the bottom of the air compressor cylinder head.

Note : Air compressor heads may have one or two spring pins depending on the part number.



Install the air compressor cylinder head gasket. The air compressor cylinder head gasket also has larger diameter holes on one side to assist in proper orientation and should be located on the spring pin protruding from the bottom of the air compressor cylinder head. When properly installed, the air compressor cylinder head gasket will allow free movement of the air compressor intake reed valve.



Note : If improperly installed, the air compressor cylinder head gasket will not allow proper movement of the intake reed valve and will prevent the air compressor from building air pressure properly.

Install

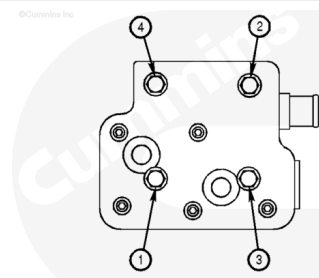
Note : When installing a used air compressor cylinder head, you must use new gaskets and capscrews. These components are not reusable and are available in the gaskets and seals kit. See the Assembly procedure above for proper assembly of the necessary components.

Install the four air compressor head mounting capscrews.

Tighten the capscrews in the sequence shown. Use the torque plus angle method.

Torque Value:

1. 25 n•m [221 in-lb]
2. Rotate 90 degrees



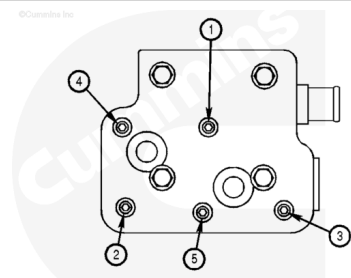
⚠ CAUTION ⚠

If the proper torque sequence is not followed as directed, probable failure of the air compressor head may result.

Tighten the five Torx™ head capscrews in the sequence shown.

Torque Value:

1. 6 n•m [53 in-lb]
2. Rotate 90 degrees.



Finishing Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Install the air inlet and outlet connections to the air compressor. Refer to Procedure 012-014 in Section 12. (</qs3/pubsys2/xml/en/procedures/56/56-012-014-tr.html>)
- Install the air compressor governor signal line. See equipment manufacturer service information.
- Fill the cooling system. Refer to Procedure 008-018 in Section 8. (</qs3/pubsys2/xml/en/procedures/56/56-008-018-tr.html>)
- Connect the batteries. See equipment manufacturer service information.
- Operate the engine. Check for leaks and proper operation.

Last Modified: 11-Oct-2022
